

4.17 Quiz: Early Finishers

4. [Maximum mark: 5]

Events A and B are such that $P(A) = 0.4$, $P(A|B) = 0.25$ and $P(A \cup B) = 0.55$.

Find $P(B)$.

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Do **not** write solutions on this page.

8. [Maximum mark: 17]

In a large city, 160 people were surveyed. Of those, 60 were children (C) and the rest adults (A).

Each person in the survey was asked whether they preferred milk chocolate (M) or dark chocolate (D). It was found that 48 of the children preferred milk chocolate. This information is shown in the following table.

	M (milk chocolate)	D (dark chocolate)	Total
C (children)	48	p	60
A (adults)	x	y	q

- (a) Find the value of
- (i) p ;
 - (ii) q . [2]
- (b) Three people are chosen at random from those surveyed. Find the probability that all three are adults. [4]
- (c) (i) Given that $P(A|M) = \frac{1}{3}$, find the value of x .
- (ii) A person is chosen at random from those surveyed. Write down the probability that they are an adult who prefers milk chocolate. [4]
- (d) Determine if the events A and M are independent. Justify your answer. [3]

It can be assumed that the survey results are representative of the population of the city.

- (e) Ten people in the city are chosen at random. Find the probability that at least five of them prefer dark chocolate. [4]

